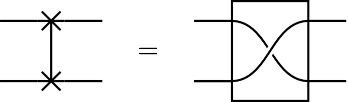

$$\Longleftrightarrow \quad \hat{S}^{[0,1]} = \sum_{j,k=0}^{d-1} |j\rangle\langle k| \otimes |k\rangle\langle j|$$